BIVARIANT CHOW COHOMOLOGY OF A SCHEME OF FINITE TYPE OVER A FIELD WITH TARGET THE POINT IS ORDINARY CHOW HOMOLOGY IN COMPLEMENTARY DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBW (sources/stacks-chow/pinned-original-source.tex line 14183).

Allowed standard prerequisites (editorial).

- Bivariant operational Chow groups A^p(X -> Y) and their axioms, including the empty-class reduction. Same chapter, chow.tex, Lemma labelled lemma-bivariant-zero (source line 6585), invoked at proof line 14236. Supplies that it suffices to test equality of bivariant classes against the classes of integral schemes; the construction of the class is inside the counted span.
- Flat pullback and proper pushforward are bivariant operations. Same chapter, Lemmas labelled lemma-flat-pullback-bivariant (source line 6131) and lemma-push-proper-bivariant (source line 6167), invoked at proof lines 14214 and 14222. Supply the two operations whose composite g_* o nu o f^* realises the desired class; the identification of that composite with c_alpha and the inverse check are in the counted span.
- Writing a cycle class as a finite sum of integral closed subschemes. Same chapter, Definition labelled definition-cycles (source line 1535), invoked at proof line 14206. Supplies the representation alpha = sum n_i [X_i] that the construction of the bivariant class starts from.

Source reference. Original definition 02QR: sources/stacks-chow/pinned-original-source.tex, source lines 1535–1535. Same-chapter pointer into the author environment definition at source line 1535; invoked at proof line 14206 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma

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at source line 6131; invoked at proof line 14214 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 14222 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 14236 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let X be a scheme locally of finite type over k. Then we have a canonical identification*

$$A^p(X \to \mathrm{Spec}(k)) = \mathrm{CH}_{-p}(X)$$

for all $p \in \mathbf{Z}$.

Proof. Consider the element $[\mathrm{Spec}(k)] \in \mathrm{CH}_0(\mathrm{Spec}(k))$. We get a map $A^p(X \to \mathrm{Spec}(k)) \to \mathrm{CH}_{-p}(X)$ by sending c to $c \cap [\mathrm{Spec}(k)]$.

Conversely, suppose we have $\alpha \in \mathrm{CH}_{-p}(X)$. Then we can define $c_\alpha \in A^p(X \to \mathrm{Spec}(k))$ as follows: given $X' \to \mathrm{Spec}(k)$ and $\alpha' \in \mathrm{CH}_n(X')$ we let

$$c_\alpha \cap \alpha' = \alpha \times \alpha'$$

in $\mathrm{CH}_{n-p}(X \times_k X')$. To show that this is a bivariant class we write $\alpha = \sum n_i [X_i]$ as in Definition 02QR. Consider the composition

$$\coprod X_i \xrightarrow{g} X \to \mathrm{Spec}(k)$$

and denote $f : \coprod X_i \to \mathrm{Spec}(k)$ the composition. Then g is proper and f is flat of relative dimension $-p$. Pullback along f is a bivariant class $f^* \in A^p(\coprod X_i \to \mathrm{Spec}(k))$ by Lemma 0B78. Denote $\nu \in A^0(\coprod X_i)$ the bivariant class which multiplies a cycle by n_i on the ith component. Thus $\nu \circ f^* \in A^p(\coprod X_i \to X)$. Finally, we have a bivariant class

$$g_* \circ \nu \circ f^*$$

by Lemma 0EPK. The reader easily verifies that c_α is equal to this class and hence is itself a bivariant class.

To finish the proof we have to show that the two constructions are mutually inverse. Since $c_\alpha \cap [\mathrm{Spec}(k)] = \alpha$ this is clear for one of the two directions. For the other, let $c \in A^p(X \to \mathrm{Spec}(k))$ and set $\alpha = c \cap [\mathrm{Spec}(k)]$. It suffices to prove that

$$c \cap [X'] = c_\alpha \cap [X']$$

when X' is an integral scheme locally of finite type over $\mathrm{Spec}(k)$, see Lemma 02UC. However, then $p' : X' \to \mathrm{Spec}(k)$ is flat of relative dimension $\dim(X')$ and hence $[X'] = (p')^*[\mathrm{Spec}(k)]$. Thus the fact that the bivariant classes c and c_α agree on $[\mathrm{Spec}(k)]$ implies they agree when capped against $[X']$ and the proof is complete. □